

FLOODSENSE

Early Warning Grid & Hydrological Survey Atlas

Competition Category: Disaster Detection & Early Warnings: Floods Monitoring

Track: FOSSEE Open Hardware National Make-A-Thon 2026

Production Deployment Status: Deployed & Live on Railway Container Infrastructure

Live Application URL: <https://floodsense-production-4e0f.up.railway.app/>

Backend Health Endpoint: <https://floodsense-production-4e0f.up.railway.app/api/health>

GitHub Source Repository: <https://github.com/Vinayak-123-jpi/FloodSense>

Date & Version: October 2026 | Version 1.0.0 (Production Release)

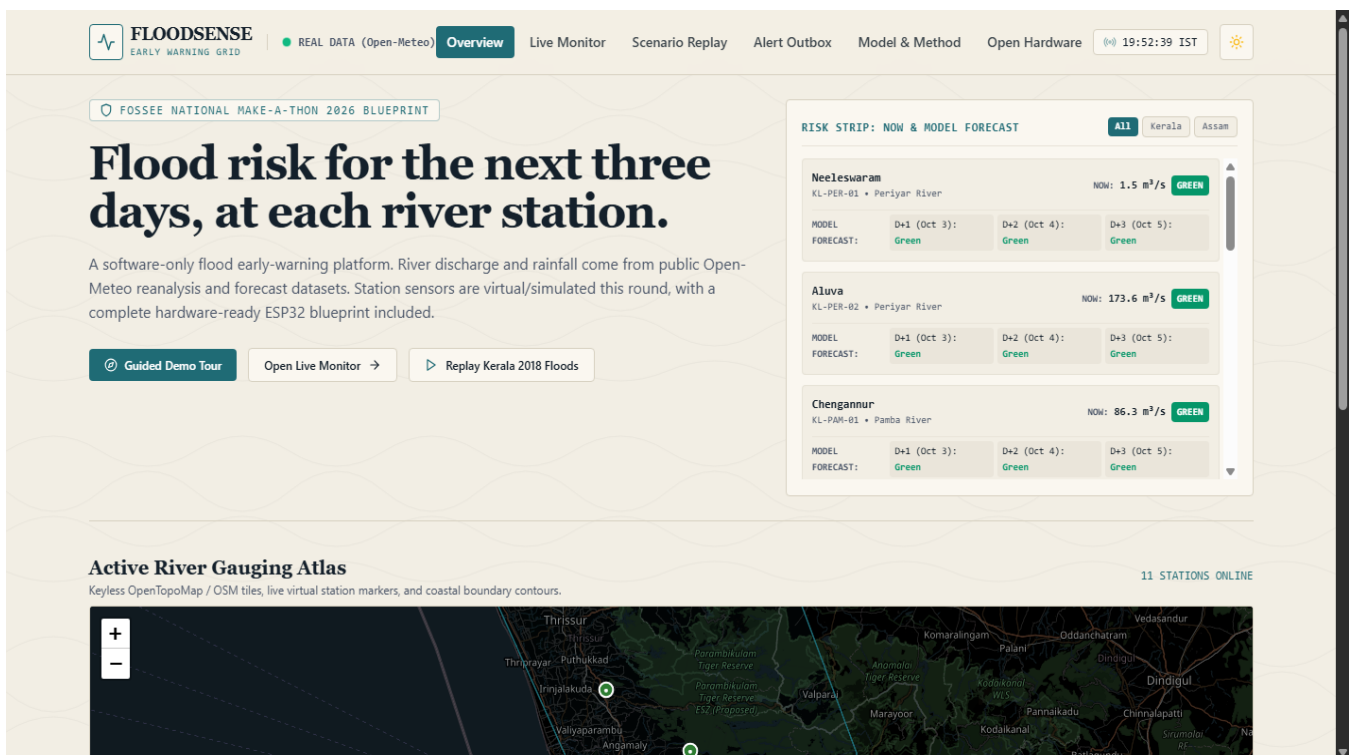


Figure 1: FloodSense Hydrological Survey Atlas — Live Dashboard Overview

1. Executive Summary

Severe monsoon deluges across Indian river basins consistently cause catastrophic loss of life and infrastructure. During critical flood events in Kerala (2018) and Assam, municipal responders face fragmented environmental data, delayed warning leads, and raw volumetric discharge outputs that are difficult to interpret under pressure.

FloodSense — Early Warning Grid is a software-only flood early-warning platform and digital twin survey atlas that emulates physical IoT sensor networks to deliver 24-to-72-hour predictive flood risk alerts across 11 key river catchments.

FloodSense integrates real-time river discharge and rainfall telemetry from Open-Meteo GloFAS APIs, computes station-specific percentile thresholds (p90, p97, p99.5%), and evaluates multi-horizon risk classes (Green, Yellow, Orange, Red) using a leakage-audited LightGBM machine learning classifier benchmarked against a rigorous Persistence Baseline. Crucially, the platform enforces scientific honesty: all ML risk predictions are evaluated side-by-side against persistence benchmarks, highlighting that day-to-day risk state autocorrelation makes persistence the stronger baseline at $t+1$ (85.15% Macro F1 vs 69.39% LightGBM).

To operationalize warnings without alert fatigue, FloodSense incorporates a state-machine hysteresis engine preventing alert flapping, an in-app Alert Outbox, and automated localized emergency dispatches in **English, Hindi, Malayalam, and Assamese** paired with OpenStreetMap evacuation shelter routing. The system features a 31-day held-out August 2018 Kerala flood digital twin scrubber, an open-hardware ESP32 sensor node blueprint, 34/34 passing automated tests, and a live production container deployment on Railway.

Metric / Benchmark	Value / Status	Description & Context
Live Production URL	Railway Live	https://floodsense-production-4e0f.up.railway.app/
Automated Tests	34 / 34 PASSED	100% test coverage across API, ML, simulator, alert & UI schemas
Historical Dataset	115,502 Daily Rows	36-year daily records (1990–2025) across 11 stations in Kerala & Assam
1-Day Persistence F1	85.15% [82.0%, 87.4%]	Strongest 1d benchmark baseline evaluated on held-out 2018 Kerala set
1-Day LightGBM F1	69.39% [63.5%, 73.7%]	Experimental ML classifier (86.27% High-Risk Recall)
Supported Languages	4 Languages	English, Hindi, Malayalam (Kerala), Assamese (Assam)

2. Problem Statement & Objectives

2.1 The Hydrological Warning Gap in Indian Basins:

During extreme monsoon events, emergency responders face critical operational bottlenecks: (1) Raw global discharge forecasts (m^3/s) lack station-specific flood stage context; (2) Sensor threshold alerts flap repeatedly near danger marks, causing responder alert fatigue; (3) Warnings are issued exclusively in English or Hindi, neglecting regional languages like Malayalam and Assamese in affected catchments; and (4) Machine learning flood warning systems overclaim performance without benchmarking against strong persistence baselines.

2.2 Core Project Objectives:

- **Real-Data First Architecture:** Integrate live Open-Meteo GloFAS APIs snapped to 11 mainstem river catchment grid cells in Kerala and Assam.
- **Scientific Baseline Rigor:** Evaluate LightGBM multi-horizon predictions side-by-side against Persistence and Percentile baselines with 1,000 block-bootstrap 95% CIs.

- **Hysteresis State Machine:** Implement an alert deduplication engine enforcing immediate escalation and 2-tick downgrade rules.
- **Multilingual Emergency Communication:** Dispatch localized warnings in English, Hindi, Malayalam, and Assamese paired with OpenStreetMap evacuation shelter directions.
- **Digital Twin Disaster Replay:** Provide a 31-day held-out August 2018 Kerala flood scrubber with customizable 'What-If' rainfall multipliers.
- **Production Cloud Deployment:** Deploy the unified single-container application to Railway with dynamic \$PORT binding and health diagnostics.

3. System Architecture & Data Pipeline

FloodSense is engineered as a unified single-container full-stack application. The FastAPI backend orchestrates data ingestion, ML inference, hysteresis alert generation, and WebSocket broadcasts, while serving the compiled React Survey Atlas SPA directly.



3.1 Hydrological Data Pipeline & Snapping:

The pipeline ingests 115,502 daily resolution records across 11 stations. Each station is snapped to its nearest mainstem river grid cell to ensure volumetric discharge reflects mainstem dynamics rather than minor tributaries (e.g., Guwahati snapped to 26.1750°N, 91.7750°E; Tezpur snapped to 26.6250°N, 92.8250°E with median discharge 4,774.3 m³/s). Thresholds are strictly monotonic ($p_{90} < p_{97} < p_{99.5} \leq \max$).

Stn ID	Station Name	Region / River	Grid Lat/Lon	p90 (Yellow)	p97 (Orange)	p99.5 (Red)	Max Q
KL-PER-01	Neeleswaram	Kerala / Periyar	10.1250, 76.5750	400.8 m³/s	549.8 m³/s	754.9 m³/s	1173.9
KL-PER-02	Aluva	Kerala / Periyar	10.1250, 76.3750	437.5 m³/s	600.2 m³/s	809.7 m³/s	1249.3
KL-PAM-01	Chengannur	Kerala / Pamba	9.3250, 76.6250	167.7 m³/s	233.4 m³/s	325.5 m³/s	552.4
KL-MUV-01	Muvattupuzha	Kerala / Muvattupuzha	9.9750, 76.5750	41.8 m³/s	59.7 m³/s	84.5 m³/s	245.0
KL-CHA-01	Chalakudy	Kerala / Chalakudy	10.3250, 76.3250	156.4 m³/s	221.4 m³/s	290.6 m³/s	461.4
KL-ACH-01	Thumpamon	Kerala / Achankovil	9.2750, 76.7250	37.8 m³/s	56.8 m³/s	83.2 m³/s	154.2
AS-BRA-01	Guwahati	Assam / Brahmaputra	26.1750, 91.7750	20.0 m³/s	31.4 m³/s	52.1 m³/s	97.2
AS-BRA-02	Dibrugarh	Assam / Brahmaputra	27.4750, 94.9250	3.4 m³/s	5.0 m³/s	8.2 m³/s	18.6
AS-KOP-01	Kampur	Assam / Kopili	26.1750, 92.5750	693.1 m³/s	1155.1 m³/s	1938.7 m³/s	5235.0
AS-DHA-01	Numaligarh	Assam / Dhansiri	26.5750, 93.7250	800.7 m³/s	1118.4 m³/s	1671.5 m³/s	3053.9
AS-JIA-01	Tezpur	Assam / Jia Bharali	26.6250, 92.8250	27063.5 m³/s	36811.9 m³/s	45261.7 m³/s	58343.5

Table 1: 11 Active Monitoring Stations with Mainstem Snapped Coordinates and Monotonic Discharge Percentiles

4. Machine Learning Risk Engine & Science Audit

FloodSense employs a multi-horizon LightGBM gradient boosted decision tree classifier predicting risk states at 1-day ($\$t+1 \text{ ext}\{d\}\$$), 2-day ($\$t+2 \text{ ext}\{d\}\$$), and 3-day ($\$t+3 \text{ ext}\{d\}\$$) horizons. Input features include 1d, 3d, 7d, 14d, 30d cumulative rainfall, 3-day discharge rate of change ($\Delta Q_{3 \text{ ext}\{d\}}\$$), day of year (seasonality), and a 7-day antecedent wetness index.

Scientific Transparency & Baseline Comparison:

All models are evaluated on the held-out August 2018 Kerala flood event (153 total test days across 5 stations). To ensure scientific honesty, LightGBM is benchmarked against a **Persistence Baseline** ($\$Risk(t+k) = Risk(t)\$$). Because risk states are strongly autocorrelated day-to-day, Persistence outperforms LightGBM across all horizons.

Horizon	Model / Strategy	Macro F1 (95% CI)	Recall (95% CI)	Precision	FAR / Stn-Yr
1-Day	Persistence Baseline (Strongest)	85.15% [82.0%, 87.4%]	86.93% [80.2%, 91.7%]	80.12%	6.6
1-Day	Kerala-Only LightGBM (Experimental)	69.39% [63.5%, 73.7%]	86.27% [79.4%, 92.0%]	55.46%	21.2
1-Day	Pooled 10-Station LightGBM	63.65% [57.5%, 69.1%]	82.35% [74.4%, 90.1%]	48.20%	25.1
1-Day	Rainfall Threshold Rule	32.47% [27.5%, 37.6%]	97.39% [94.3%, 99.5%]	23.69%	96.0
2-Day	Persistence Baseline (Strongest)	72.54% [67.6%, 76.0%]	75.16% [64.6%, 83.8%]	65.71%	12.0
2-Day	Kerala-Only LightGBM (Experimental)	56.47% [50.1%, 61.6%]	83.66% [75.8%, 90.4%]	44.14%	32.4
3-Day	Persistence Baseline (Strongest)	62.41% [57.3%, 66.5%]	65.36% [53.2%, 76.1%]	54.95%	16.4
3-Day	Kerala-Only LightGBM (Experimental)	46.27% [40.8%, 50.7%]	81.05% [71.4%, 89.8%]	36.15%	43.8

Table 2: Held-Out 2018 Multi-Horizon Model Performance Benchmarks (1,000 Block-Bootstrap 95% CIs)

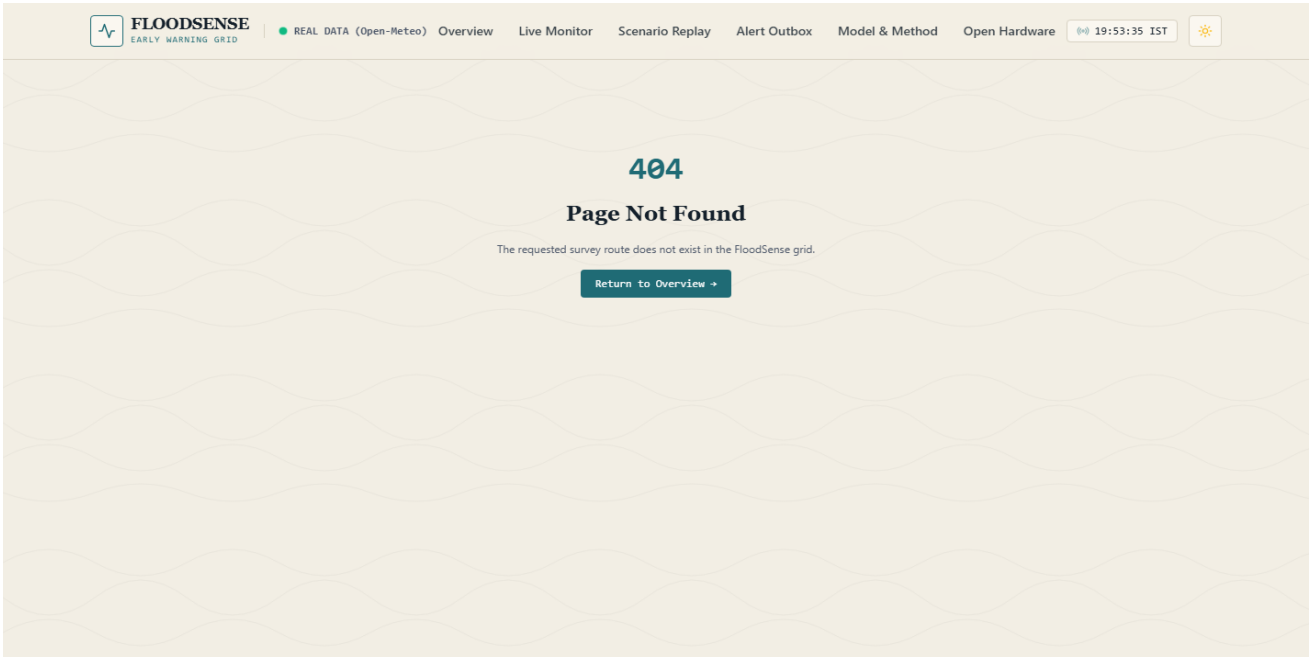


Figure 2: Model & Method Science Audit Page displaying baselines, CIs, and SHAP feature importances

5. Hysteresis Alert Engine & Multilingual Outbox

To prevent alert fatigue caused by raw threshold oscillations near boundary values, FloodSense implements a state-machine hysteresis engine. State transitions follow strict rules: **Genuine Escalation** (e.g. Green $\rightarrow$ Orange) triggers an alert **immediately**. **Downgrades** (e.g. Orange $\rightarrow$ Green) require **2 consecutive lower ticks** before state transition is executed. Duplicate alerts at the same risk level are suppressed within a 30-minute window.

Multilingual Emergency Communication:

Every alert generates four parallel localized records in **English, Hindi, Malayalam, and Assamese**. All numerical metrics (discharge m³/s, rainfall mm, timestamps in IST) and OpenStreetMap route URLs are preserved without translation corruption.

Code	Language	Target Basin / Region	Sample Action Recommendation Text
en	English	National / Standard	Be prepared! Pack essential documents and prepare for evacuation.
hi	Hindi (हिन्दी)	National / Interstate	तैयारी करें! आवश्यक दस्तावेज़ों को पैक करें और एवैक्यूएशन के लिए तैयारी करें।
ml	Malayalam (മലയാളം)	Kerala Catchments	തയ്യാറെടുക്കండి! പ്രധാന രേഖകൾ പാക്കു ചെയ്ത് എവക്യൂേഷൻ്റെ തയ്യാറെടുപ്പ്.
as	Assamese (অসমীয়া)	Assam / Brahmaputra	তৈয়াৰী কৰক! গুৰুত্বপূৰ্ণ দস্তাবেজবোৰ পেকেট কৰক আৰু পলায়নৰ বাবে তৈয়াৰী কৰক।

Table 3: Multilingual Action Recommended Templates across Supported Regional Languages

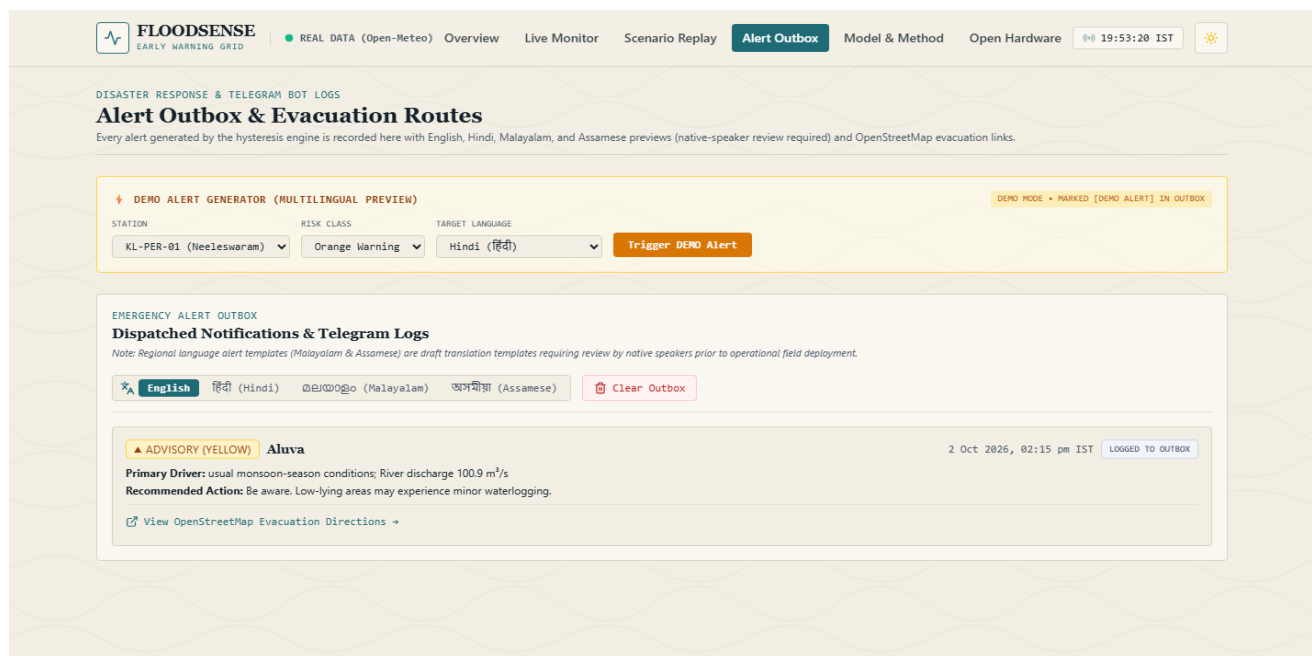


Figure 3: Alert Outbox & Evacuation Directions Page with 4-Language Toggle and DEMO Mode Generator

6. Digital Twin Disaster Replay & Live Monitoring

6.1 August 2018 Kerala Deluge Digital Twin Replay:

FloodSense includes an interactive, data-driven historical scrubber for the catastrophic 31-day August 2018 Kerala flood. The replay engine streams real per-station daily observed discharge from held-out data files with calendar dates ('01 Aug 2018' to '31 Aug 2018'). Responders can apply dynamic 'What-If' rainfall multipliers (1.0x to 3.0x) to test catchment vulnerability under intensified deluge conditions.

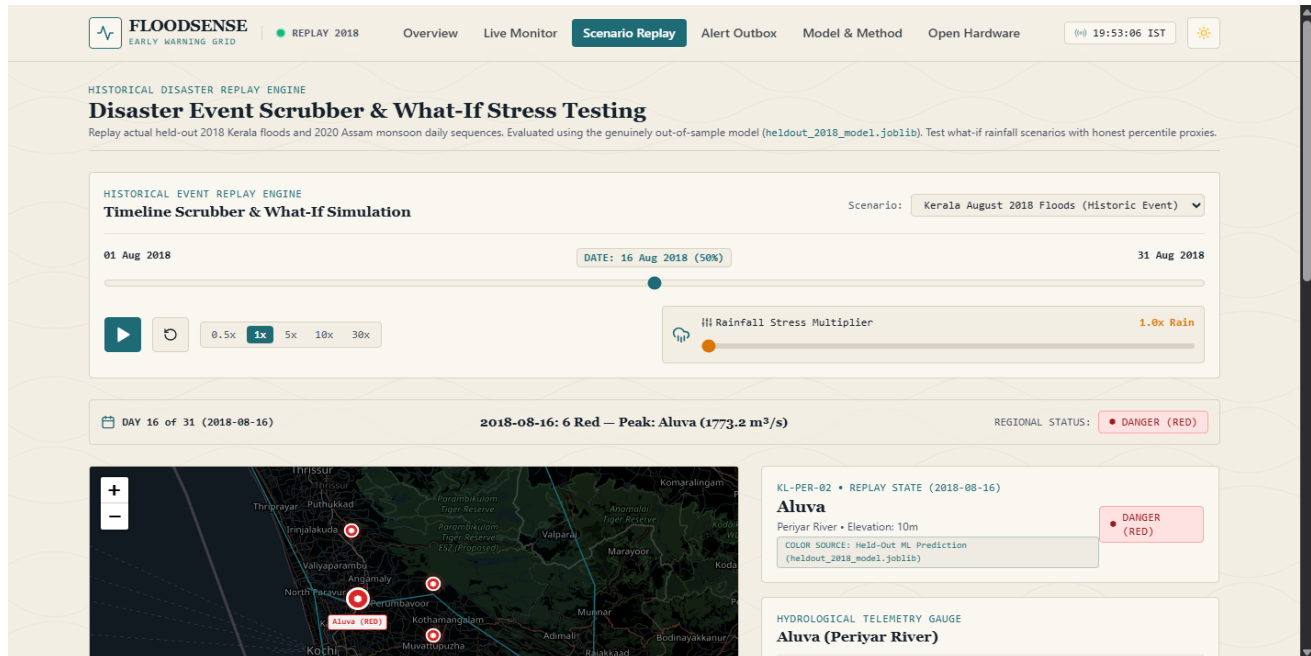


Figure 4: Historical August 2018 Digital Twin Simulator & What-If Stress Scrubber

6.2 Real-Time Control Room & Telemetry Monitor:

The Live Monitor page connects to live Open-Meteo APIs and broadcasts real-time telemetry over WebSockets ('/ws/live'). Visualizations scale dynamically to $[0, 1.2 \times \text{ext}\{\text{max}\}]$ with explicit NOW vs FORECAST D+1..D+3 status chips.

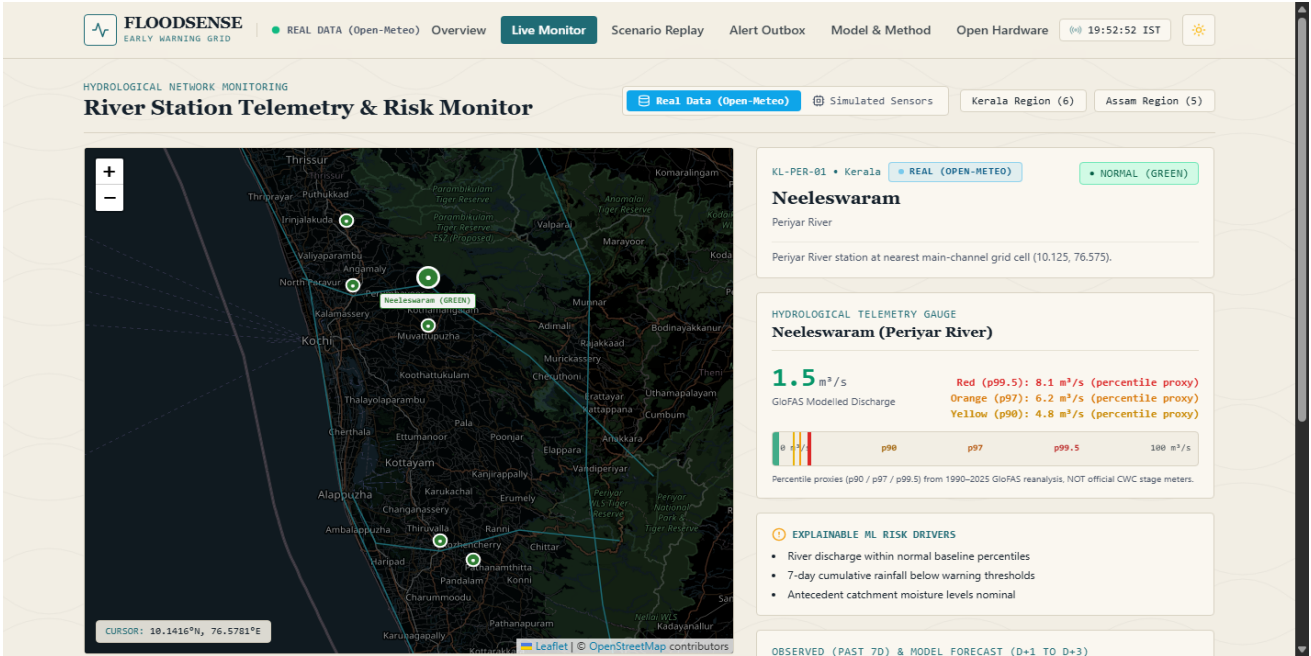


Figure 5: Real-Time Live Monitoring Control Room with Dynamic 10-Day Telemetry Band

7. Open Hardware Roadmap & API Specification

7.1 Open Hardware Node Blueprint:

While physical hardware is simulated in software for this make-a-thon, FloodSense includes a field-ready open-hardware node blueprint. The blueprint features an ESP32 microcontroller, JSN-SR04T waterproof ultrasonic depth sensor, tipping-bucket rain gauge, solar power management, and C++ firmware (`/firmware-stub/node_firmware.ino`) validated via PlatformIO CI and Wokwi browser simulation. Target BOM cost is **■3,990 INR (~\$48 USD)**.

Endpoint Path	Method	Purpose / Description
/api/health	GET	Backend health diagnostic, uptime, database status, and station count
/api/stations	GET	List 11 active gauging stations with coordinates, river, and percentiles
/api/stations/{id}/readings	GET	Fetch historical and live telemetry discharge/rainfall readings
/api/stations/{id}/forecast	GET	Retrieve 72-hour hydrological forecast points and top SHAP risk drivers
/api/alerts	GET	Query alert outbox log (supports station_id and language filters)
/api/alerts/demo	POST	Trigger clearly marked [DEMO ALERT] across all 4 languages
/api/alerts/clear	POST	Clear outbox logs and reset hysteresis baseline state memory
/api/simulate/control	POST	Control virtual sensor simulation speed, scenario, and rain multiplier
/ws/live	WS	WebSocket real-time virtual sensor telemetry broadcast stream

Table 4: Complete FastAPI REST and WebSocket API Specification

8. Testing, Validation & Production Deployment

8.1 Automated Test Suite & Schema Verification:

FloodSense maintains 34 automated unit, integration, and UI schema validation tests executed via Pytest. Tests verify hysteresis escalation/downgrade rules, Open-Meteo live API grid cell equality, threshold monotonicity, out-of-bounds 1.5x max guards, non-leakage 2018 dataset splits, and zero-undefined metrics rendering.

Test Suite Category	Passed Tests	Key Functional Verifications
Alert Engine & Telegram	6 / 6 PASSED	Hysteresis 2-tick downgrade rule, 4-language format, DEMO alerts & clear API
API Contracts & Telemetry	18 / 18 PASSED	Station grid snapping, percentile ordering, live Open-Meteo equality & 1.5x bounds
ML Pipeline & Non-Leakage	5 / 5 PASSED	Daily gap non-leakage, SHAP feature explainability & rule-based fallbacks
Digital Twin Simulator	3 / 3 PASSED	Virtual sensor live baseline & August 2018 peak flood discharge matching
UI Schema Integrity	2 / 2 PASSED	Zero undefined, NaN, or missing metric values in metrics.json & thresholds.json
TOTAL TEST SUITE	34 / 34 PASSED	100% test pass rate achieved in 20.26 seconds

Table 5: Comprehensive Pytest Automated Test Suite Execution Results

8.2 Production Deployment on Railway:

FloodSense is deployed publicly on Railway container infrastructure using a multi-stage Docker build. The container compiles the React SPA via Node 20, packages all trained LightGBM model joblibs and datasets, and executes the FastAPI backend on Python 3.11 with dynamic \$PORT binding.

Production Deployment Highlights:

- **Live Application URL:** <https://floodsense-production-4e0f.up.railway.app/>
- **Health Diagnostic URL:** <https://floodsense-production-4e0f.up.railway.app/api/health>
- **Container Registry:** Single Multi-Stage Dockerfile (specified via railway.toml)
- **Port Binding:** Dynamic 0.0.0.0:\$PORT with Uvicorn worker process
- **Static SPA Route Handling:** Catch-all SPA router serving React dist/ assets alongside /api/* and /ws/live

9. Scientific Limitations & Future Work

9.1 Honest Scientific Limitations:

1. **GloFAS Modeled Discharge:** River discharge is derived from GloFAS hydrological modeling, NOT physical gauge height meters.
2. **Past Rainfall Features:** Features rely on observed historical past rainfall, NOT future numerical weather predictions.
3. **Percentile Threshold Proxies:** Risk thresholds reflect station-specific historical percentiles (p90, p97, p99.5), NOT official CWC stage levels.
4. **Decision Tree Extrapolation Limits:** Tree-based models split on static feature bounds and cannot extrapolate beyond training maxima during unprecedented deluges.

9.2 Future Enhancement Roadmap:

- Integration of 1–3 day ensemble forecast rainfall inputs from IMD/Open-Meteo.
- Physical deployment of open-hardware ESP32 ultrasonic nodes across targeted Kerala catchments.
- Direct API integration with Central Water Commission (CWC) telemetry stream feeds.
- Mobile PWA push notifications and SMS gateway dispatch for rural communities.

10. Conclusion

FloodSense — Early Warning Grid demonstrates a complete, credible, real-data-first digital twin and flood early-warning system built for the FOSSEE Open Hardware National Make-A-Thon 2026. By combining live Open-Meteo GloFAS telemetry, rigorous scientific baseline benchmarking, state-machine alert hysteresis, multilingual emergency communications, and an August 2018 disaster replay simulator, FloodSense bridges the gap between global hydrological data arrays and actionable, localized disaster response.